Improving the Efficiency of the PC Algorithm by Using Model-Based Conditional Independence Tests

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What are constraint-based algorithms and PC?

Constraint-based algorithms infer a structure that corresponds to the conditional independence (CI) relationships within a population. These conditional independence relationships are inferred by performing CI tests on a data sample.

The PC algorithm [1] is a constraint-based algorithm with the following steps:
1. Initialize a complete undirected graph *G*.
2. For each pair of variables, perform CI tests in order of increasing size of conditioning sets *(0, 1, 2, …)* and stop only when a test infers that the pair of variables is CI (in this case, remove the edge corresponding to the pair in *G*) or after all feasible CI tests have been performed.
3. Use the information from those tests to orient the edge directions and return the final directed acyclic graph (DAG) *G*.

Worst-case runtime of the PC algorithm

The phase of PC and of other constraint-based algorithms that applies CI tests typically dominates their runtime. The number of CI tests applied in the worst case is equal to the cardinality of the power set of the possible conditioning variables.

Why use model-based CI tests?

Most constraint-based algorithms assume that CI tests perform more accurately on small conditioning sets. However, recent model-based CI tests [2,3,4] have the potential to work effectively on very large cond. sets because the models typically used in those tests are well-regularized. We present a pre-processing algorithm for PC that utilizes large cond. sets.

P3PC: Pre-processing Plus PC

P3PC(X, V): X is a dataset and V is a set of variables
1. Initialize p as an $n \times n$ matrix with all 1 values
2. Initialize L as a list of sets having length c_1
3. for a, b in V:
   4. if $a \perp b$, set $p[a,b] = p[b,a] = 0$; continue
   5. for $i \in \{1, \dots, c_1\}$, set $L[i]$ as a random set of $n - c_2$ vars from $V \backslash \{a, b\}$
   6. for $S \in L$, if $a \perp b | S$, set $p[a,b] = p[b,a] = 0$; continue
7. Return *PC(X,V,p)* $p[i.j] = 0$ if vars i,j could be made CI; 1 o/w

Preliminaries

A trail is a sequence of distinct edges.
A collider of a trail is at the head of the two adjacent edges.
A trail is blocked by a cond. set if at least one node on the trail that is not a collider is in the cond. set or if at least one node on the trail that is a collider is not in the cond. set.
A pair of variables could be made CI if all trails between them are blocked by a cond. set.

An Erdős-Renyi (ER) DAG on n nodes is randomly generated as follows: the n nodes are labeled in increasing order and an edge exists with probability p and can only have a direction from nodes with a smaller label to nodes with a larger label.

Theoretical analysis on Erdős-Renyi DAGs

We show that if a pair of variables is CI, then for large conditioning sets of size $n-4$, the probability that all trails between the pair will be blocked is high in three statements:
Statement 1 (Trails of length at least 7): A conditioning set of size $n-4$ will block every trail of length 7 or greater.
Statement 2 (Trails of length at most 6): In an Erdős-Renyi network, the expected number of trails of length ≤ 6 between a pair of nodes is at most $p(1 + (pn) + (pn)^2 + \dots + (pn)^6) \leq 7n^6p^7$ assuming that $p \geq 1/n$. If $p = \Theta(1)/n$, the expectation tends to 0 as $n \to \infty$.
Statement 3 (Expected number of colliders): The expected number of nodes that are colliders for some trail is $n(1-p)^n - (1-p)^n + (2((1-p)^n - 1)/p + n - p^2 + 1$. For example, when $p = 1/n$, the expression corresponds to $0.104n$.

Empirical analysis on Erdős-Renyi DAGs and on DAGs that correspond to real-world phenomena

We performed experiments to estimate the raw number of CI tests performed by P3PC and by PC. The experiments use known DAG structure and standard d-separation rules to determine if variables could be made CI by a given cond. set, rather than running CI tests on data generated by that DAG.

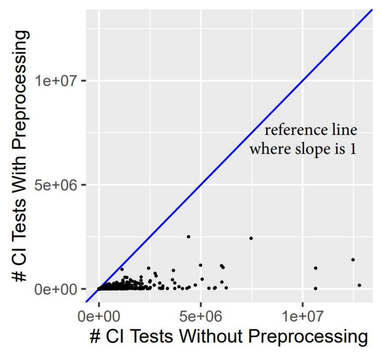
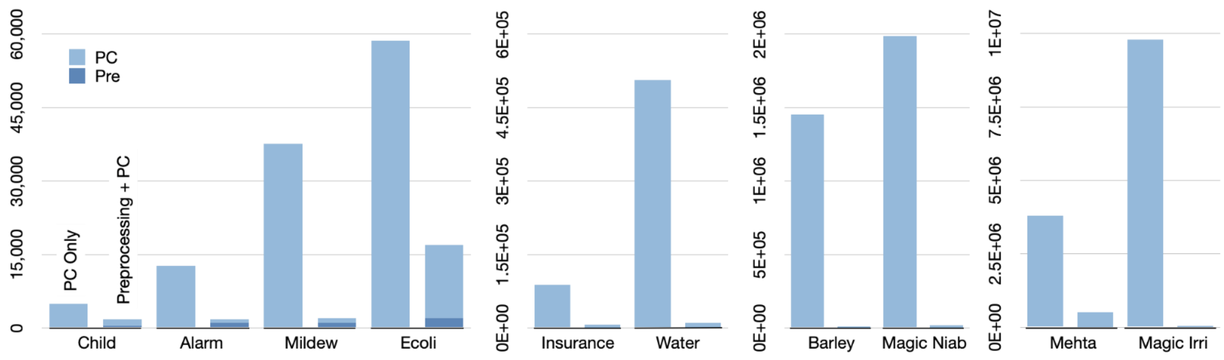


Figure 1. The number of tests performed by the PC algorithm with pre-processing (P3PC) including the number of tests performed in the pre-processing step on the y-axis and the number of tests performed by the PC algorithm without pre-processing on the x-axis. Each point in the plot corresponds to a randomly generated ER DAG. The blue line is for reference and has a slope of 1.

Table 1. The number of nodes and edges for each DAG from the *bnlearn* repository that corresponds to real-world systems, and the proportion of tests performed by P3PC to test performed by the PC algorithm alone for each.

	Child	Alarm	Mildew	Ecoli	Insurance	Water	Barley	Magic Niab	Mehra	Magic Irri
# Nodes	20	37	35	46	27	32	48	44	24	64
# Edges	25	46	46	70	52	66	84	66	71	102
Prop. Tests	.361	.143	.055	.285	.073	.021	.006	.008	.137	.005

Figure 2. For DAGs that correspond to real-world systems in the *bnlearn* repository, the number of tests that P3PC performs to learn a structure and that the PC algorithm alone performs to learn a structure.



Takeaways about P3PC

1. We present P3PC, which performs model-based CI tests for variables pairs on large conditioning sets first.
2. We show that P3PC reduces the total number of CI tests performed by the PC algorithm in expectation for ER DAGs.
3. We perform experiments on DAGs that correspond to real-world systems, finding that P3PC performs between *0.5%* to *36%*, and often less than *10%*, the number of tests performed by PC alone. We find similar surprising results on ER DAGs.
4. These results imply a new reason to prioritize research on accurate model-based tests of conditional independence, particularly those that perform well with large conditioning sets.

References

[1] Peter Spirtes, Clark Glymour, and Richard Scheines. *Causation, Prediction, and Search*, volume 81. 01 1993.
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